Physical Account (Annex C): PoseidonOS, Team 6

Every number the software relies on that is not in Annex A/B, or that narrows an Annex value. All 113 assumptions are in `data/assumptions/*.yaml` with value, unit, range, reason and source, and are read through `A("<id>")`. CI fails on any other numeric literal in `shipos/physics` or `shipos/engine`. The **appendix** (generated, pages 2+) lists every use site by file and line.

Class	What we assumed	Value / range	Why	Where used in code
Electrical power	Reserve at stopped/slow = cruise (not credited higher)	200 / 100 kW (2 / 1 gensets)	A.10: judge on the lower of slow and cruise. Scenario B quotes ~100 kW at 3 kn	`physics/power.py` `reserve_kw`, `engine/kernel.py`
	Reserve at fast	150 / 75 kW	bottom of A.1 range; outside default envelope	`physics/power.py`
	Line lost on trip	stbd line, bus-tie to port	A.9 leaves it to us	`physics/power.py` `surviving_lines`
	Spinning reserve held back when degraded	0 kW	A.9 ~100 kW is already net of hotel load	`engine/shedding.py` `_capacities`
	Overdraw warn / throttle / shed	+10 % / +15 % after 10 s / shed after 60 s	brief: guests exceed by 10–20 %	`engine/shedding.py` overdraw guard
Heat rejection	kW electric → kW heat	1.0 (A.2)	stated so it is traceable	`physics/cooling.py` `heat_kw`
	Loop vs seawater	linear 160 kW @12 °C → 120 kW @30 °C, clamped; ×0.6 degraded (72 kW @30 °C)	A.2 "falling linearly"	`physics/cooling.py` `loop_capacity_kw`
	Inertia (+30 kW / 60 s / 10 min)	modelled, **not** used by D4	conservative: D4 never counts on it	`physics/cooling.py` `InertiaBuffer`
Positions	Bay 1 half mass limit	4 t each	8 t bay split evenly	`data/ship`, `physics/positions.py`
	Hangar module / roof limits	1.2 t; roof 30 kg, 0.5 m², 0.5 m antenna height	G4 hangar+drone 1.04 t; roof takes G5 antenna only	`physics/positions.py`
	Free mast slot is next to the nav radar	true → G1 –30 %	B.G1	`physics/positions.py` `capability_factor`
	One item per mounting point	1	A.3 "two 10-ft guests" = two halves	`physics/positions.py` `check`
Mass / stability	Margin kept beyond A.4	0 t (full 10 t and 12 t-eq)	no hidden safety factor	structural: `physics/mass.py` uses the A.4 values directly
	G2 split body / winch / probe	3.0 / 0.4 / 0.1 t	card gives 3.5 t total	`data/guests/G2.yaml`
Data links	Ship's own SATCOM share	0.5 Mbit/s	weather, ops, crew	`physics/links.py`
	SATCOM admission basis	2 Mbit/s × 70 % – 0.5 = 0.9 Mbit/s	worst case; elastic, so loss is reported, not a refusal	`physics/links.py` `capacity_mbps`
	Acoustic drop	down if manoeuvring, deck machinery or layered water	A.5	`physics/links.py` `acoustic_up`
State transitions	Plan on A.6 upper bound; speed/course/noise settle concurrently	e.g. quiet 4 min; deck clear 5 / 12 min (14 / 6 crew)	a deadline met on paper is met at sea	`statemachine.py` planner
	Speed band edges	0.5 / 7.5 / 16 kn	midpoints of A.6 gaps	`physics/power.py` `speed_band`
Compute · Response time	Per-guest cores (G1–G7)	4 / 2 / 8+1 acc / 8+1 acc / 4 / 4 / 2	per-card workload	`physics/compute.py`
	D4 horizon / first action / step	30 min / t+1 s / 2 s steps	ceilings 5 s / 60 s	`engine/shedding.py`
	Solver budgets	≤ 200 000 placements, ≤ 3 restrictions, ≤ 20 000 shed combos, then deterministic fallback	stay under 30 s / 5 s	`engine/placement.py`, `admission.py`, `shedding.py`
Priorities · Crew	Capability weights high / medium / low	1.0 / 0.5 / 0.2; mission-critical ≥ 0.8	B.0: resolved from priority file	`data/priorities/*.yaml`, all engines
	Confirmation delay	15–60 s (14), 60–180 s (6), forbidden at 0	A.6	`crew/policy.py`
	Six-crew handling window	07:00–19:00 local	reduced crew, daylight only	`sim/env.py`
Anything else	G3 recovery 15 min, suspended leg restarts after 75 min; G4 approach/launch 3 min, top-up 30 min	as stated	card gives totals only	`data/guests/G3.yaml`, `G4.yaml`
	G6 (our card): towed acoustic array 2.5 t; winch 15 kW × 20 min deploy/recover; listening 4 kW at 4–6 kn, quiet; 120-min transect, 30-min pause; port-only 3 people / 4 h	17 invented values	civil marine-mammal monitoring	`data/guests/G6.yaml`, `data/assumptions/g6.yaml`
	G7: survivor-care module 2.4 t, 3 / 18 kW + 10 kW × 20 s peak, requires G5, safety-critical	invented, ranges in registry	exercises B.0 flags	`data/guests/G7.yaml`
	Environment: Open-Meteo Strait of Sicily cache, hourly hold, sea-state from Hs (WMO 3700)	cache ≤ 24 h old	offline, replayable	`sim/env.py`, `data/env_cache/`

Brief inconsistency recorded. Annex B says "any four exceed the 10 t margin". By the card masses, G1+G2+G4+G5 = 7.61 t and G1+G3+G4+G5 = 9.61 t fit. We follow the arithmetic (`tests/physics/test_mass_positions.py`).

Closing question: one change to the platform. Add **40 kW of chilled-water capacity at 30 °C seawater (120 → 160 kW)**, for example a second plate exchanger on the existing loop. In warm water the loop, not the gensets, is what pushes a degraded ship into shedding. A loop degraded by 40 % at 30 °C gives 72 kW for guests, while one genset still gives 100 kW, so cooling binds first, and there is no cooling repair at sea (A.6). With +40 kW the degraded loop holds 96 kW, within 4 kW of the one-genset reserve. In Scenario A with G4 aboard, the worst-case heat shortfall (G5 45 + G4 35 + G2 20 = 100 kW) falls from 28 kW to 4 kW, so one throttle step covers it instead of a shed list. This proposal comes from our model and has not been validated on a real ship.

Physical Account: assumption registry

Generated by `python -m tools.physical_account` from `data/assumptions/*.yaml`; do not edit. Narrative: PHYSICAL_ACCOUNT.md.

A *use* is a literal `A("<id>")` call in `shipos/` or `tools/` (Python) or a `"$<id>"` value in `data/` outside `data/assumptions/` (YAML). Tests and docs are not counted.

113 assumptions: 104 used, 9 unused; 1 dynamic lookup(s).

Dynamic lookups

Calls whose id is computed at run time; the ids they reach are not resolved here.

- `shipos/data.py:18` `A(node[len(REF):])`

Unused

No literal `A()` call or `$` reference reads these; some record modelling choices the code makes structurally.

- `data.satcom_admission_basis` (defined `data/assumptions/data_links.yaml:8`)
- `environment.sea_state_proxy` (defined `data/assumptions/environment.yaml:27`)
- `heat.capacity_interpolation` (defined `data/assumptions/heat.yaml:8`)
- `mass.margin_kept_t` (defined `data/assumptions/mass.yaml:3`)
- `positions.bay1_sharing` (defined `data/assumptions/positions.yaml:33`)
- `power.bus_tie_transfer` (defined `data/assumptions/power.yaml:50`)
- `state.plan_on_upper_bound` (defined `data/assumptions/state_transitions.yaml:18`)
- `state.transitions_concurrent` (defined `data/assumptions/state_transitions.yaml:23`)
- `state.crew_level_lookup` (defined `data/assumptions/state_transitions.yaml:28`)

Compute · Response time (`data/assumptions/compute_time.yaml`)

id	value	unit	range	why	source	defined	used at
`compute.g1.cores`	4	cores	—	Signal processing for a passive RF sensor.	invented	`data/assumptions/compute_time.yaml:3`	`data/guests/G1.yaml:16`
`compute.g1.mem_gb`	8	GB	—	See compute.g1.cores.	invented	`data/assumptions/compute_time.yaml:4`	`data/guests/G1.yaml:16`
`compute.g2.cores`	2	cores	—	Profile processing and buffering.	invented	`data/assumptions/compute_time.yaml:5`	`data/guests/G2.yaml:34`
`compute.g2.mem_gb`	4	GB	—	See compute.g2.cores.	invented	`data/assumptions/compute_time.yaml:6`	`data/guests/G2.yaml:34`
`compute.g3.cores`	8	cores	—	Sonar and imagery post-processing on the cradle.	invented	`data/assumptions/compute_time.yaml:7`	`data/guests/G3.yaml:43`
`compute.g3.mem_gb`	32	GB	—	See compute.g3.cores.	invented	`data/assumptions/compute_time.yaml:8`	`data/guests/G3.yaml:43`
`compute.g3.accelerators`	1	accelerators	—	Imagery inference for cable / pipeline inspection.	invented	`data/assumptions/compute_time.yaml:9`	`data/guests/G3.yaml:43`
`compute.g4.cores`	8	cores	—	Video ingest and detection.	invented	`data/assumptions/compute_time.yaml:10`	`data/guests/G4.yaml:41`
`compute.g4.mem_gb`	16	GB	—	See compute.g4.cores.	invented	`data/assumptions/compute_time.yaml:11`	`data/guests/G4.yaml:41`
`compute.g4.accelerators`	1	accelerators	—	Person-in-water detection on drone video.	invented	`data/assumptions/compute_time.yaml:12`	`data/guests/G4.yaml:41`
`compute.g5.cores`	4	cores	—	The node brings its own edge compute; the rack only hosts its control plane.	invented	`data/assumptions/compute_time.yaml:13`	`data/guests/G5.yaml:23`
`compute.g5.mem_gb`	16	GB	—	See compute.g5.cores.	invented	`data/assumptions/compute_time.yaml:14`	`data/guests/G5.yaml:23`
`compute.g7.cores`	2	cores	—	Patient-monitor aggregation and telemedicine relay.	invented	`data/assumptions/compute_time.yaml:15`	`data/guests/G7.yaml:24`
`compute.g7.mem_gb`	8	GB	—	See compute.g7.cores.	invented	`data/assumptions/compute_time.yaml:16`	`data/guests/G7.yaml:24`
`placement.margin_tie_tolerance`	0.01	normalised margin	—	Placements whose minimum margins differ by less than 1 % are treated as equal, so the flexibility and preference criteria decide instead of noise.	invented	`data/assumptions/compute_time.yaml:17`	`shipos/engine/placement.py:206`
`placement.max_candidates`	200000	assignments	—	Solver budget guard. The five-card problem has fewer than 100 assignments; above this limit the search refuses rather than blowing the 30 s ceiling.	invented	`data/assumptions/compute_time.yaml:22`	`shipos/engine/placement.py:247`
`placement.alternatives_reported`	3	options	—	D2 must show the alternatives considered; three are enough for an operator to see why the choice won.	invented	`data/assumptions/compute_time.yaml:27`	`shipos/engine/placement.py:328`
`admission.max_restrictions`	3	actions	—	Fix sets for one admission are searched up to three throttle / standby / off / interlock actions. Beyond that the answer is a displacement, and the search stays well under the	invented	`data/assumptions/compute_time.yaml:32`	`shipos/engine/admission.py:147`

id	value	unit	range	why	source	defined	used at
				30 s ceiling.			
admission.alternatives_reported	4	options	—	The ranked yes-if list shows the chosen fix plus up to four costlier ones.	invented	data/assumptions/compute_time.yaml:37	shipos/engine/admission.py:403 shipos/engine/admission.py:490
time.s_per_min	60	s/min	—	Unit conversion.	definition	data/assumptions/compute_time.yaml:42	shipos/engine/admission.py:284 shipos/engine/arbitration.py:59 shipos/engine/shedding.py:134 shipos/engine/shedding.py:168 shipos/engine/shedding.py:211 shipos/engine/shedding.py:234 shipos/engine/shedding.py:363 shipos/engine/shedding.py:376 shipos/engine/shedding.py:382 shipos/engine/shedding.py:393
time.min_per_h	60	min/h	—	Unit conversion.	definition	data/assumptions/compute_time.yaml:47	shipos/engine/admission.py:282
time.s_per_h	3600	s/h	—	Unit conversion.	definition	data/assumptions/compute_time.yaml:52	shipos/engine/admission.py:260
time.min_per_day	1440	min/day	—	Unit conversion. Losing an asset costs its weight × the mission minutes left without it.	definition	data/assumptions/compute_time.yaml:57	shipos/engine/arbitration.py:115
units.percent	100	%	—	Unit conversion for battery percentages.	definition	data/assumptions/compute_time.yaml:62	shipos/engine/arbitration.py:69 shipos/engine/shedding.py:399
display.minute_decimals	1	decimals	—	Minutes are reported to 0.1 min in decisions.	invented	data/assumptions/compute_time.yaml:67	shipos/engine/arbitration.py:198 shipos/engine/arbitration.py:209 shipos/engine/arbitration.py:257 shipos/engine/arbitration.py:282 shipos/engine/arbitration.py:295 shipos/engine/arbitration.py:304 shipos/engine/shedding.py:376 shipos/engine/shedding.py:393
shed.horizon_min	30	min	—	D4 forecasts the guests' scheduled modes 30 min ahead, so an overlap that has not started yet (a 60-min non-interruptible recharge, a 20-min drone charge) is resolved before it trips the last genset.	invented	data/assumptions/compute_time.yaml:72	shipos/engine/shedding.py:234 shipos/engine/shedding.py:363 shipos/sim/execution.py:857
shed.first_action_s	1	s	—	The first shed command leaves 1 s after the trip is detected (A.8 ceiling 5 s).	invented	data/assumptions/compute_time.yaml:77	shipos/engine/shedding.py:314 shipos/engine/shedding.py:466
shed.step_interval_s	2	s	—	Each later step waits 2 s so the remaining genset settles between steps. Up to 30 steps still fit in the 60 s full-sequence ceiling.	invented	data/assumptions/compute_time.yaml:82	shipos/engine/shedding.py:314
shed.min_relief_kw	0.1	kW	—	Avoids dividing by zero when ranking an armed step that relieves nothing at the current peak.	invented	data/assumptions/compute_time.yaml:87	shipos/engine/shedding.py:257
display.kw_decimals	2	decimals	—	kW figures in decisions are reported to 0.01 kW.	invented	data/assumptions/compute_time.yaml:92	shipos/engine/shedding.py:140 shipos/engine/shedding.py:150 shipos/engine/shedding.py:375 shipos/engine/shedding.py:439 shipos/engine/shedding.py:465
shed.max_defer_entries	2	schedule entries per guest	—	Solver budget on the 5 s path. Only a guest's next two deferrable transitions inside the horizon are candidates for deferral.	invented	data/assumptions/compute_time.yaml:97	shipos/engine/shedding.py:200
shed.max_defer_delays	4	delays per entry	—	Solver budget. A deferral is tried up to the next four other-guest transitions (the instants where the overlap can end).	invented	data/assumptions/compute_time.yaml:102	shipos/engine/shedding.py:202
shed.max_combinations	20000	combinations	—	Count-based cap that keeps D4 deterministic and under the 5 s first-action ceiling. Above it the planner falls back to adding one move per guest in the declared ladder order.	invented	data/assumptions/compute_time.yaml:107	shipos/engine/shedding.py:290 shipos/engine/shedding.py:292

Data links (`data/assumptions/data_links.yaml`)

id	value	unit	range	why	source	defined	used at
data.ship_satcom_reserved_mbps	0.5	Mbit/s	—	The ship's own traffic (weather, ops, crew) on the shared SATCOM; guests share what is left.	invented	data/assumptions/data_links.yaml:3	data/ship/reference_vessel.yaml:63

id	value	unit	range	why	source	defined	used at
`data.satcom_admission_basis`	worst case (2 Mbit/s × 70 % availability)	rule	—	Admission is judged on the envelope; SATCOM is elastic, so a shortfall is reported as capability lost, not a refusal.	invented	`data/assumptions/data_links.yaml:8`	**unused**

Environment (`data/assumptions/environment.yaml`)

id	value	unit	range	why	source	defined	used at
`environment.sample_hold_s`	3600	s	[3600, 3600]	Hourly samples are held constant only between consecutive known hourly endpoints. No interpolation across missing hours or extension past the final sample.	B3 demonstration approximation; Open-Meteo hourly APIs	`data/assumptions/environment.yaml:3`	`shipos/sim/env.py:14`
`environment.cache_max_age_s`	86400	s	[0, 172800]	Reject a forecast downloaded more than one day before the explicit scenario start unless archived replay is explicitly requested. This is a demo freshness policy, not a provider guarantee.	B3 demonstration policy	`data/assumptions/environment.yaml:9`	`shipos/sim/env.py:15`
`environment.daylight_start_hour`	7	local hour	[7, 7]	PLAN B3 restricts physical handling with six crew to local 07:00-19:00. This clock rule is not astronomical daylight.	docs/PLAN.md section 4 B3	`data/assumptions/environment.yaml:15`	`shipos/sim/env.py:16`
`environment.daylight_end_hour`	19	local hour	[19, 19]	Six-crew operations must finish by 19:00 local; finishing exactly at 19:00 is allowed.	docs/PLAN.md section 4 B3	`data/assumptions/environment.yaml:21`	`shipos/sim/env.py:17`
`environment.sea_state_proxy`	WMO 3700 height guidance applied to model significant wave height; exact bounds use lower code	convention	—	The contract takes a sea-state code while Open-Meteo supplies significant wave height. This proxy does not represent a visual sea-state observation or maximum individual wave height.	https://www.nodc.noaa.gov/gtspp/document/codetbls/wmocodes/table3700.html	`data/assumptions/environment.yaml:27`	**unused**

G6 towed passive acoustic array (`data/assumptions/g6.yaml`)

id	value	unit	range	why	source	defined	used at
`guests.g6.body_mass_t`	2.5	t	[1.5, 3.5]	Cable drum with tow winch, ~1 km of tow cable, array and recorder rack in one deck module.	invented (docs/PLAN.md B1 recommendation)	`data/assumptions/g6.yaml:5`	`data/guests/G6.yaml:7`
`guests.g6.listening_kw`	4	kW	[2, 6]	Hydrophone pre-amplifiers, digitiser and recorder rack while the array is streamed.	invented (docs/PLAN.md B1 recommendation)	`data/assumptions/g6.yaml:11`	`data/guests/G6.yaml:26`
`guests.g6.standby_kw`	0.5	kW	[0.2, 1]	Recorder rack idle and drum brake held; also the draw while the array is streamed but not recording.	invented	`data/assumptions/g6.yaml:17`	`data/guests/G6.yaml:12` `data/guests/G6.yaml:34`
`guests.g6.winch_kw`	15	kW	[10, 20]	Tow winch paying out or hauling in the cable over the stern ramp; below G2's 20 kW CTD winch because the load is lighter.	invented	`data/assumptions/g6.yaml:23`	`data/guests/G6.yaml:14` `data/guests/G6.yaml:36`
`guests.g6.deploy_min`	20	min	[15, 30]	Stream the array and pay out the tow cable to working length at tow speed.	invented	`data/assumptions/g6.yaml:29`	`data/guests/G6.yaml:15` `data/guests/G6.yaml:45`
`guests.g6.recover_min`	20	min	[15, 30]	Haul in the cable and recover the array up the ramp; same length as deployment.	invented	`data/assumptions/g6.yaml:35`	`data/guests/G6.yaml:37`
`guests.g6.transect_min`	120	min	[60, 240]	One listening transect long enough for a useful marine-mammal and ambient-noise record.	invented	`data/assumptions/g6.yaml:41`	`data/guests/G6.yaml:27` `data/guests/G6.yaml:48`

id	value	unit	range	why	source	defined	used at
guests.g6.tow_speed_lo_kn	4	kn	[3, 5]	Below ~4 kn the array sinks out of its depth band and the tow cable can foul.	invented	data/assumptions/g6.yaml:47	data/guests/G6.yaml:21 data/guests/G6.yaml:30
guests.g6.tow_speed_hi_kn	6	kn	[5, 8]	Above ~6 kn flow noise on the hydrophones drowns the signal.	invented	data/assumptions/g6.yaml:53	data/guests/G6.yaml:21 data/guests/G6.yaml:30
guests.g6.ramp_sea_state_max	4	sea state	[3, 5]	Handling the array over the ramp; matches the sea-state limit Annex B gives G2 and G3 for over-side work.	invented (by analogy with B.G2 / B.G3)	data/assumptions/g6.yaml:59	data/guests/G6.yaml:23
guests.g6.pause_window_min	30	min	[15, 45]	The array can stay streamed, not recording, for this long while the ship manoeuvres; longer and the operator recovers it.	invented	data/assumptions/g6.yaml:65	data/guests/G6.yaml:50
guests.g6.backbone_mbps	20	Mbit/s	[10, 50]	Multi-channel raw hydrophone stream to the recorder and the edge compute.	invented	data/assumptions/g6.yaml:71	data/guests/G6.yaml:54
guests.g6.satcom_mbps	0.2	Mbit/s	[0.1, 0.5]	Detection summaries ashore only; raw audio stays aboard, so the demand is elastic.	invented	data/assumptions/g6.yaml:77	data/guests/G6.yaml:55
guests.g6.cores	4	cores	[2, 8]	Spectrogram and click / whistle detection on CPU; no accelerator needed.	invented	data/assumptions/g6.yaml:83	data/guests/G6.yaml:56
guests.g6.mem_gb	16	GB	[8, 32]	Buffering several minutes of multi-channel audio for detection.	invented	data/assumptions/g6.yaml:89	data/guests/G6.yaml:56
guests.g6.install_people	3	people	[2, 4]	Lift the module into the bay and lead the tow cable to the ramp.	invented	data/assumptions/g6.yaml:95	data/guests/G6.yaml:62
guests.g6.install_hours	4	h	[3, 6]	Module lift, manifold and power hook-up, cable lead and a dry test.	invented	data/assumptions/g6.yaml:101	data/guests/G6.yaml:62

Heat rejection (`data/assumptions/heat.yaml`)

id	value	unit	range	why	source	defined	used at
heat.kw_per_kw_electric	1.0	kW heat / kW electric	—	Annex A.2 states 1:1. Listed so the Physical Account shows it is used, not because we changed it.	A.2	data/assumptions/heat.yaml:3	shipos/physics/cooling.py:22
heat.capacity_interpolation	linear	model	—	A.2 gives 160 kW at 12 °C and 120 kW at 30 °C, "falling linearly". We clamp outside 12-30 °C to the end values (no extrapolation).	A.2 + clamp invented	data/assumptions/heat.yaml:8	**unused**

Mass / stability (`data/assumptions/mass.yaml`)

id	value	unit	range	why	source	defined	used at
mass.margin_kept_t	0	t	—	We use the full 10 t margin and the 12 t-eq stability budget from A.4, with no extra safety factor; admission reports the remaining margin.	A.4	data/assumptions/mass.yaml:3	**unused**
guests.g2.body_mass_t	3.0	t	—	B.G2 gives 3.5 t including the winch and probe parts; we split it 3.0 / 0.4 / 0.1.	invented split of B.G2	data/assumptions/mass.yaml:8	data/guests/G2.yaml:4
guests.g2.winch_mass_t	0.4	t	—	See guests.g2.body_mass_t.	invented split of B.G2	data/assumptions/mass.yaml:13	data/guests/G2.yaml:9
guests.g2.probe_mass_t	0.1	t	—	See guests.g2.body_mass_t.	invented split of B.G2	data/assumptions/mass.yaml:18	data/guests/G2.yaml:10

Mission priorities · Crew (`data/assumptions/mission_crew.yaml`)

id	value	unit	range	why	source	defined	used at
priority.level.high	1.0	weight	—	Capability levels in the cards (high / medium / low) map to weights; high is the unit.	invented	data/assumptions/mission_crew.yaml:3	data/priorities/patrol.yaml:5 data/priorities/sar_primary.yaml:3 data/priorities/survey_primary.yaml:5
priority.level.medium	0.5	weight	—	Half of high.	invented	data/assumptions/mission_crew.yaml:8	data/priorities/patrol.yaml:5 data/priorities/sar_primary.yaml:3 data/priorities/survey_primary.yaml:5
priority.level.low	0.2	weight	—	Low but not zero, so a low-priority guest still beats nothing.	invented	data/assumptions/mission_crew.yaml:13	data/priorities/patrol.yaml:5 data/priorities/sar_primary.yaml:3 data/priorities/survey_primary.yaml:5
priority.mission_critical_threshold	0.8	weight	—	Mission-critical is derived from the priority file (B.0), not a stored boolean; only "high" guests qualify.	invented	data/assumptions/mission_crew.yaml:18	data/priorities/patrol.yaml:7 data/priorities/sar_primary.yaml:5

id	value	unit	range	why	source	defined	used at
							data/priorities/survey_primary.yaml:7
admission.interlock_capability_fraction	0.05	fraction of a guest's capability	—	An interlock defers one short operation (e.g. a 5-min winch run) for at most one charging cycle (≤ 20 min), a few times a day. We price that at 5 % of the deferred guest's capability.	invented	data/assumptions/mission_crew.yaml:23	shipos/engine/admission.py:116
arbitration.horizon_min	60	min	—	D3 counts the minutes a winner's follow-on activity (e.g. the drone's next launch) stays blocked over the next hour. That is about one survey leg plus recovery, and long enough to separate "pause and re-run" from "suspend".	invented	data/assumptions/mission_crew.yaml:28	shipos/engine/arbitration.py:114

Anything else (data/assumptions/other.yaml)

id	value	unit	range	why	source	defined	used at
guests.g2.standby_kw	1.0	kW	—	The card gives no standby draw; logger and buffers stay alive.	invented	data/assumptions/other.yaml:3	data/guests/G2.yaml:24
guests.g3.survey_speed_lo_kn	2.5	kn	—	Card says surveying at 3 kn; we allow ±0.5 kn.	invented	data/assumptions/other.yaml:8	data/guests/G3.yaml:15
guests.g3.survey_speed_hi_kn	3.5	kn	—	See guests.g3.survey_speed_lo_kn.	invented	data/assumptions/other.yaml:13	data/guests/G3.yaml:15
guests.g3.recovery_min	15	min	—	Recovering the vehicle up the ramp; the card only gives 45 min for a full swap.	invented	data/assumptions/other.yaml:18	data/guests/G3.yaml:22
guests.g3.suspend_window_min	75	min	—	A suspended leg can only restart after recovery (15 min) and the non-interruptible 60-min recharge.	derived from B.G3 + guests.g3.recovery_min	data/assumptions/other.yaml:23	data/guests/G3.yaml:39
guests.g4.topping_min	30	min	—	The card says 35 kW for 20 min "then 10 kW"; we end the 10 kW phase after 30 min.	invented	data/assumptions/other.yaml:28	data/guests/G4.yaml:30
guests.g4.landing_approach_min	3	min	—	Final approach and touchdown once the ship is in the flight-deck state.	invented	data/assumptions/other.yaml:33	data/guests/G4.yaml:34
guests.g4.launch_min	3	min	—	Same as the approach.	invented	data/assumptions/other.yaml:38	data/guests/G4.yaml:25 data/guests/G4.yaml:35
guests.g5.antenna_height_m	0.4	m	—	A 0.3 m² comms antenna; used for the hangar-roof drone-clearance check.	invented	data/assumptions/other.yaml:43	data/guests/G5.yaml:10
guests.g7.mass_t	2.4	t	[1.8, 3.0]	A 10-ft medical container (two stretchers, warming, oxygen) weighs 1.8–3 t; fits a Bay 1 half (≤ 4 t) or Bay 2.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:48	data/guests/G7.yaml:6
guests.g7.standby_kw	3	kW	[2, 4]	Climate control and charged medical equipment kept ready while no one is aboard the module.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:54	data/guests/G7.yaml:11
guests.g7.treating_kw	18	kW	[12, 25]	Two patients: forced-air warming 2 × 3 kW, oxygen concentrators 2 × 1.5 kW, HVAC 6 kW, monitors 3 kW.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:60	data/guests/G7.yaml:13
guests.g7.compressor_peak_kw	10	kW	[5, 15]	Inrush when the HVAC and oxygen compressors start; exercises the A.1 peak rule.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:66	data/guests/G7.yaml:14
guests.g7.compressor_peak_s	20	s	[10, 30]	Compressor start transient; well inside the A.1 60 s peak tolerance.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:72	data/guests/G7.yaml:15
guests.g7.satcom_mbps	1	Mbit/s	[0.5, 2]	Telemedicine video at low resolution plus vital-sign telemetry.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:78	data/guests/G7.yaml:22
guests.g7.backbone_mbps	50	Mbit/s	[10, 100]	Monitor feeds to the bridge and the edge node.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:84	data/guests/G7.yaml:23
guests.g7.install_people	4	people	[3, 4]	Same crane-and-hook-up crew as the other bay guests (B.G2, B.G5).	invented (G7 is a team-6 card)	data/assumptions/other.yaml:90	data/guests/G7.yaml:30
guests.g7.install_h	4	h	[4, 8]	Lower end of A.11's 4–8 h per bay guest: plug-in container, no winch to commission.	invented (G7 is a team-6 card)	data/assumptions/other.yaml:96	data/guests/G7.yaml:30

Positions (data/assumptions/positions.yaml)

id	value	unit	range	why	source	defined	used at
positions.bay1_half_max_mass_t	4	t	—	Bay 1 is ≤ 8 t as a whole; we split it evenly between its two 10-ft halves.	invented	data/assumptions/positions.yaml:3	data/ship/reference_vessel.yaml:32 data/ship/reference_vessel.yaml:34
positions.hangar_max_mass_t	1.2	t	—	The A.3 hangar takes one VTOL ≤ 150 kg; the G4 hangar module (900 kg) plus drone (140 kg) must fit, plus a little margin.	invented	data/assumptions/positions.yaml:8	data/ship/reference_vessel.yaml:38
positions.hangar_roof_max_mass_t	0.03	t	—	The hangar roof takes only a small antenna (G5's 25 kg). B.G5 names the roof as an alternative to the mast.	invented	data/assumptions/positions.yaml:13	data/ship/reference_vessel.yaml:41
positions.hangar_roof_max_area_m2	0.5	m²	—	Roughly a mast footprint (0.6 m²), slightly smaller.	invented	data/assumptions/positions.yaml:18	data/ship/reference_vessel.yaml:41

id	value	unit	range	why	source	defined	used at
`positions.hangar_roof_max_antenna_height_m`	0.5	m	—	Drone clearance check for the hangar roof; an antenna ≤ 0.5 m stays below the drone's approach path.	invented	`data/assumptions/positions.yaml:23`	`shipos/physics/positions.py:114`
`positions.mast3_near_radar`	True	bool	—	The only free mast slot sits next to the navigation radar, so G1 loses 30 % unless the radar is quiet (B.G1).	invented	`data/assumptions/positions.yaml:28`	`data/ship/reference_vessel.yaml:48`
`positions.bay1_sharing`	two 10-ft halves or one 20-ft	rule	—	A.3, if neither guest needs the bay clear. None of the five cards needs it.	A.3	`data/assumptions/positions.yaml:33`	**unused**

Electrical power (`data/assumptions/power.yaml`)

id	value	unit	range	why	source	defined	used at
`power.reserve.stopped.two_gensets`	200	kW	[150, 250]	Annex gives the reserve only at cruise; stopped/slow propulsion load is lower, but we do not credit it (conservative, same as cruise).	narrowed from A.1	`data/assumptions/power.yaml:3`	`data/ship/reference_vessel.yaml:8`
`power.reserve.stopped.one_genset`	100	kW	[90, 120]	Same as cruise with one genset down; Scenario B quotes ~100 kW at 3 kn, consistent.	narrowed from A.1 / A.9	`data/assumptions/power.yaml:9`	`data/ship/reference_vessel.yaml:8`
`power.reserve.slow.two_gensets`	200	kW	[150, 250]	Not credited above cruise (conservative); A.10 says use the lower reserve of slow/cruise.	narrowed from A.1	`data/assumptions/power.yaml:15`	`data/ship/reference_vessel.yaml:9`
`power.reserve.slow.one_genset`	100	kW	[90, 120]	Scenario B states the reserve halves to about 100 kW at 3 kn.	A.9 / Scenario B	`data/assumptions/power.yaml:21`	`data/ship/reference_vessel.yaml:9`
`power.reserve.fast.two_gensets`	150	kW	[150, 250]	Fast (>18 kn) propulsion load is highest, so take the bottom of the A.1 range. Fast is outside the default envelope.	narrowed from A.1	`data/assumptions/power.yaml:27`	`data/ship/reference_vessel.yaml:11`
`power.reserve.fast.one_genset`	75	kW	[50, 100]	Half of the fast two-genset value, by analogy with cruise (200 → 100).	invented	`data/assumptions/power.yaml:33`	`data/ship/reference_vessel.yaml:11`
`power.line_lost_on_trip`	stbd	line	—	A.9 lets us choose; we lose the stbd line with genset 2. Guests on stbd transfer to port via the bus-tie, so with one genset down the surviving line carries everything (still ≤ 150 kW).	choice per A.9	`data/assumptions/power.yaml:39`	`data/ship/reference_vessel.yaml:15`
`power.degraded_spinning_reserve_kw`	0	kW	[0, 25]	The A.9 ~100 kW is already the guests' share net of hotel load, so we do not hold back more by default. Raise it (e.g. to 20 kW ≈ the 10-20 % overdraw the brief warns of) to keep the last genset off its limit; D4 then sheds earlier.	invented (policy knob)	`data/assumptions/power.yaml:44`	`data/ship/reference_vessel.yaml:16`
`power.bus_tie_transfer`	True	bool	—	Needed for line_lost_on_trip; without a bus-tie, guests on the lost line would lose power at the trip.	invented	`data/assumptions/power.yaml:50`	**unused**
`overdraw.warn_ratio`	1.1	measured / declared	—	The brief warns that guests exceed their declaration by 10-20 %. At +10 % the governor warns.	brief (state the system reasons on)	`data/assumptions/power.yaml:55`	`shipos/engine/shedding.py:469`
`overdraw.throttle_ratio`	1.15	measured / declared	—	At +15 %, held for warn_after_s, the guest is throttled to its low-power mode (or standby).	invented	`data/assumptions/power.yaml:60`	`shipos/engine/shedding.py:450`
`overdraw.warn_after_s`	10	s	—	Ignore inrush and transients shorter than 10 s.	invented	`data/assumptions/power.yaml:65`	`shipos/engine/shedding.py:450`
`overdraw.shed_after_s`	60	s	—	An overdraw that persists for 60 s (as long as the peak allowance) is shed.	invented	`data/assumptions/power.yaml:70`	`shipos/engine/shedding.py:451`

State transitions (`data/assumptions/state_transitions.yaml`)

id	value	unit	range	why	source	defined	used at
`state.band_edge.stopped_slow_kn`	0.5	kn	—	A.6 defines stopped and slow 3-5 kn; we call anything above 0.5 kn under way.	invented	`data/assumptions/state_transitions.yaml:3`	`data/ship/reference_vessel.yaml:99` `data/ship/reference_vessel.yaml:100`
`state.band_edge.slow_cruise_kn`	7.5	kn	—	Midpoint of the 5-10 kn gap between A.6 bands.	invented	`data/assumptions/state_transitions.yaml:8`	`data/ship/reference_vessel.yaml:100` `data/ship/reference_vessel.yaml:101`
`state.band_edge.cruise_fast_kn`	16	kn	—	Midpoint of the 14-18 kn gap between A.6 bands.	invented	`data/assumptions/state_transitions.yaml:13`	`data/ship/reference_vessel.yaml:101` `data/ship/reference_vessel.yaml:102`
`state.plan_on_upper_bound`	True	rule	—	Every A.6 transition is planned at its slowest value, so a deadline that passes on paper passes at sea.	invented	`data/assumptions/state_transitions.yaml:18`	**unused**
`state.transitions_concurrent`	True	rule	—	A.6 gives one transition time per variable. We let speed, course and noise settle at the same time, so the ship is ready when the slowest one is. Exceptions - an acoustic hold command	invented	`data/assumptions/state_transitions.yaml:23`	**unused**

id	value	unit	range	why	source	defined	used at
				must arrive (≤ 60 s) before any manoeuvre, and a human confirms the deck only once it is clear.			
`state.crew_level_lookup`	next level down	rule	—	A.6 states times for 14, 6 and 0 aboard; a crew of 10 uses the 6-crew times.	invented	`data/assumptions/state_transitions.yaml:28`	**unused**